

STARKE COUNTY AP, IN, USA

WMO: 720593

Lat: 41.333N

Lon: 86.667W

Elev: 683

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.1	(c) 4.7	(d) -18.2	(e) 2.1	(f) 8.8	(g) -11.3	(h) 3.1	(i) 12.4	(j) 26.1	(k) 33.7	(l) 24.1	(m) 31.1	(n) 6.7	(o) 260	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.2	(c) 90.0	(d) 73.1	(e) 87.7	(f) 71.8	(g) 84.3	(h) 69.8	(i) 77.0	(j) 85.3	(k) 75.0	(l) 83.0	(m) 73.2	(n) 81.0	(o) 10.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	130.6	81.7	72.5	123.4	79.6	71.2	117.9	77.9	40.8	85.8	39.0	83.7	37.2	80.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.7	18.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-11.2	79.9	9.5	1.3	-18.1	80.8	-23.6	81.6	-29.0	82.3	-35.9	83.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	23.8	28.0	39.2	48.7	60.7	69.4	71.0	69.6	64.0	52.3	39.6	30.0
	DBStd	18.92	11.67	10.86	11.16	9.20	9.62	6.37	6.00	5.35	7.89	9.10	9.15	10.07
	HDD50	2990	812	618	369	133	19	0	0	0	1	81	333	624
	HDD65	6241	1277	1035	802	494	200	28	16	19	113	409	762	1086
	CDD50	2920	0	3	35	95	351	582	650	607	422	151	21	3
	CDD65	693	0	0	3	5	67	160	200	160	84	14	0	0
	CDH74	6440	0	0	19	73	694	1405	1927	1430	757	135	0	0
	CDH80	1929	0	0	1	7	205	453	619	383	236	25	0	0
Wind	WSAvg	8.0	9.7	9.5	9.6	10.3	8.8	7.0	4.9	4.4	5.4	7.8	9.0	9.3
Precipitation	PrecAvg	41.2	3.0	2.2	2.4	3.6	4.0	4.6	4.8	4.5	3.2	3.1	3.2	2.6
	PrecMax	50.4	6.0	4.8	4.9	6.7	6.6	8.0	8.9	9.6	7.9	8.8	7.2	4.7
	PrecMin	32.9	1.3	1.1	0.9	0.9	1.2	2.3	2.6	1.9	1.1	1.2	0.9	0.9
	PrecStd	5.1	1.5	1.1	1.1	1.5	2.0	2.0	1.7	2.2	2.0	1.8	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	63.0	78.8	81.2	89.8	92.6	93.5	90.3	91.1	83.7	70.1	60.8
		MCWB	53.1	46.5	62.5	61.3	70.7	72.1	76.6	73.4	73.2	66.8	57.2	54.5
	2%	DB	50.3	54.8	68.3	75.3	85.7	88.4	89.8	87.8	87.4	77.7	63.4	53.9
		MCWB	47.2	47.3	56.4	57.9	66.3	70.1	75.7	72.3	70.9	64.3	54.8	49.2
	5%	DB	45.0	49.6	62.9	70.4	81.7	85.6	86.3	84.0	81.7	72.7	58.8	47.7
		MCWB	41.7	44.5	54.3	55.9	64.0	68.4	72.9	71.6	67.3	60.3	51.4	45.0
	10%	DB	38.5	42.8	56.9	64.5	77.7	81.7	82.4	81.4	78.5	67.9	54.2	43.1
		MCWB	34.6	37.5	49.0	52.9	63.5	67.2	70.5	69.1	65.0	59.0	48.2	39.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	52.4	63.9	64.8	73.1	78.0	79.4	77.8	76.0	71.1	60.2	56.9
		MCDB	55.0	56.2	75.2	74.6	85.0	87.4	88.4	85.9	86.2	79.2	65.2	59.2
	2%	WB	47.8	48.2	60.2	61.8	70.8	75.0	77.7	75.6	73.4	66.8	56.3	49.6
		MCDB	49.8	54.0	66.1	70.4	80.1	83.2	85.7	83.3	82.4	72.1	62.3	51.6
	5%	WB	41.2	43.5	55.0	58.8	68.5	72.5	75.7	73.5	71.4	63.4	52.4	44.3
		MCDB	45.2	50.6	60.6	67.2	76.7	80.1	83.5	80.7	78.7	67.7	57.8	46.7
	10%	WB	35.4	37.9	50.3	55.3	66.2	70.7	73.7	71.8	69.1	60.3	48.4	40.2
		MCDB	37.9	41.6	57.0	63.8	73.6	77.3	81.1	78.9	74.0	67.6	54.1	43.1
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.8	18.3	21.2	20.8	20.2	21.2	21.7	22.7	20.4	16.7	13.0
		MCDBR	19.1	22.8	24.9	28.2	25.5	23.6	22.8	23.9	24.7	25.8	22.6	18.6
		MCWBR	16.5	17.9	16.1	15.7	11.4	10.1	10.4	11.2	11.1	13.2	15.2	16.4
	5% WB	MCDBR	19.3	21.4	21.6	24.2	21.6	19.1	20.0	19.7	20.7	21.3	20.0	17.3
		MCWBR	17.4	17.9	15.6	16.4	11.3	10.3	10.8	11.1	10.2	14.7	15.5	16.0
Clear-Sky Solar Irradiance	taub	0.296	0.323	0.343	0.401	0.430	0.438	0.455	0.448	0.421	0.356	0.319	0.296	
	taud	2.423	2.340	2.379	2.232	2.207	2.226	2.174	2.187	2.233	2.425	2.477	2.464	
	Ebn at Noon	273	280	287	276	270	267	261	259	257	265	262	263	
	Edh at Noon	25	32	34	43	45	44	46	44	39	29	24	22	
All-Sky Solar Radiation	RadAvg	502	768	1136	1474	1677	1920	1930	1688	1397	921	581	419	
	RadStd	48	94	91	127	153	119	103	99	114	94	75	42	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page